



Smart Textile Manufacturing Quality Inspection and Production Optimization Platform

TEST CASE DESIGN ASSESSMENT

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1. INTRODUCTION

The **Smart Textile Manufacturing Quality Inspection and Production Optimization Platform** is designed to improve the quality, efficiency, and reliability of textile manufacturing. The proposed system uses AI-based fabric defect detection, IoT sensors, production monitoring, data analytics, alerts, and digital reporting.

The purpose of this test case design is to verify that the proposed system functions correctly under normal, boundary, invalid, and exceptional conditions. Testing focuses on defect detection, production monitoring, alert generation, user management, data analysis, and reporting.

The test design applies **Equivalence Partitioning, Boundary Value Analysis, Decision Table Testing, and State Transition Testing** to improve requirement and functional coverage. The objective is to verify that the system is correct, reliable, secure, responsive, and usable.

2. FUNCTIONAL AND NON-FUNCTIONAL REQUIREMENTS

2.1 Functional Requirements

Requirement ID	Functional Requirement
FR-01	User login and authentication
FR-02	User role and access management
FR-03	Fabric image upload and acquisition
FR-04	Automatic fabric defect detection
FR-05	Defect classification by type and severity
FR-06	Defect location identification
FR-07	Real-time machine monitoring
FR-08	Production rate and downtime monitoring
FR-09	Root-cause analysis
FR-10	Production optimization recommendations
FR-11	Automated alert generation
FR-12	Quality and production report generation
FR-13	Storage and retrieval of production and inspection data
FR-14	Support for different textile manufacturing processes

2.2 Non-Functional Requirements

Requirement ID	Non-Functional Requirement
NFR-01	Performance – System should process data within an acceptable response time.
NFR-02	Scalability – System should support additional machines, users, and production lines.
NFR-03	Security – Unauthorized users should not access protected functions or data.
NFR-04	Reliability – Production and inspection data should be stored accurately.
NFR-05	Availability – Monitoring and essential services should remain accessible during operation.
NFR-06	Maintainability – System components should be easy to modify and maintain.
NFR-07	Usability – Dashboards and system functions should be clear and easy to use.

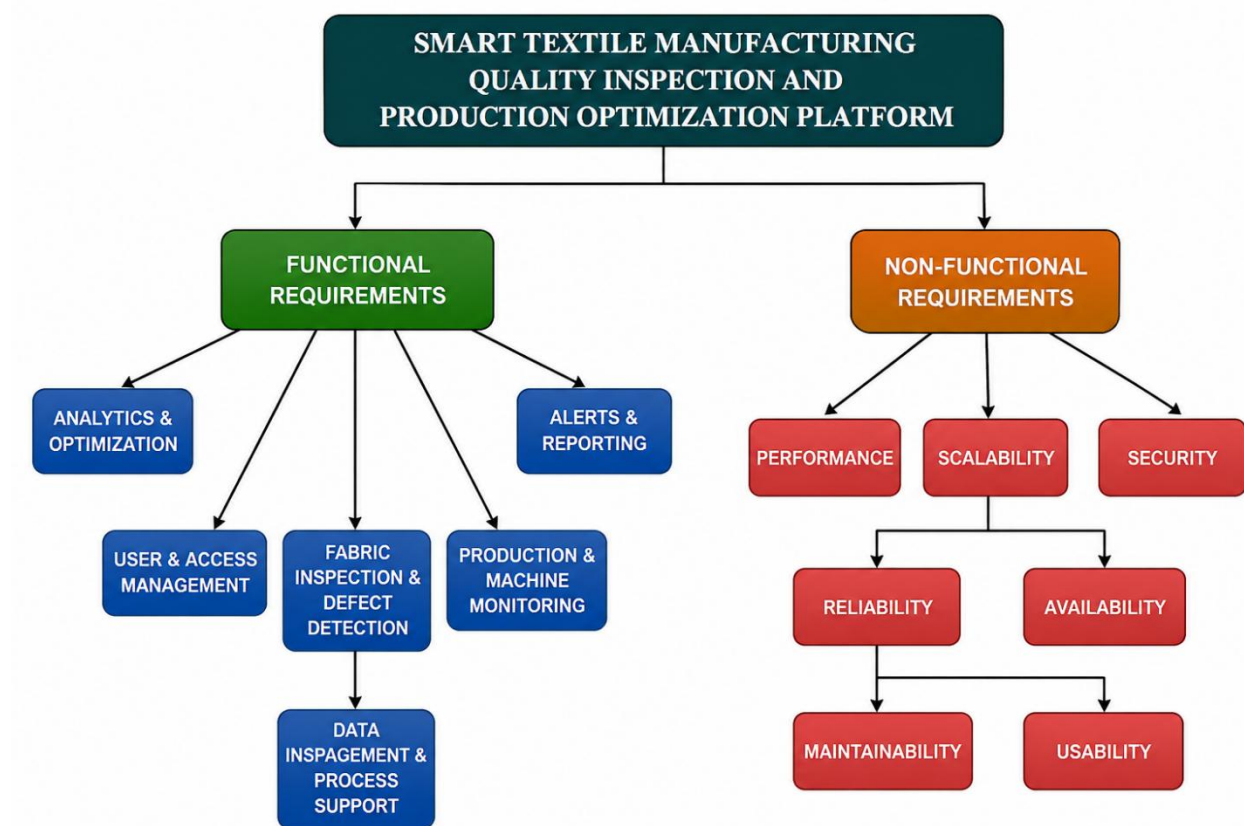


Figure 1 – Functional and Non-Functional Requirements of the Smart Textile Manufacturing Platform

3. TEST SCENARIOS

The following test scenarios are derived from the functional and non-functional requirements of the Smart Textile Manufacturing Quality Inspection and Production Optimization Platform. The scenarios cover normal, invalid, boundary, and exceptional conditions.

Scenario ID	Test Scenario	Requirement
TS-01	Verify user authentication and role-based access	FR-01, FR-02, NFR-03
TS-02	Verify valid and invalid fabric image upload	FR-03
TS-03	Verify fabric defect detection, classification, and location	FR-04, FR-05, FR-06
TS-04	Verify machine status, production rate, and downtime monitoring	FR-07, FR-08
TS-05	Verify root-cause analysis and production optimization	FR-09, FR-10
TS-06	Verify automated alert generation at configured thresholds	FR-11
TS-07	Verify quality and production report generation	FR-12, FR-13
TS-08	Verify system behavior during sensor, database, or AI service failure	NFR-04, NFR-05, NFR-06
TS-09	Verify system performance and usability under normal operation	NFR-01, NFR-07
TS-10	Verify system behavior when workload and number of connected devices increase	NFR-02

4. TEST CASES

The following test cases are designed to verify the major functions and quality requirements of the proposed platform. The test cases include normal, invalid, boundary, and exceptional conditions.

4.1 Authentication and Access Control

TC-01 – User Authentication and Access Control

Field	Details
Test Scenario	Verify login and role-based access
Test Steps	Enter valid credentials → Login → Open an authorized function → Attempt a restricted function
Test Data	Valid manager/operator credentials
Expected Result	Valid user is logged in. Authorized functions are accessible and restricted functions are denied.
Actual Result	To be recorded
Pass/Fail Status	Not Executed

4.2 Fabric Image Upload

TC-02 – Valid and Invalid Fabric Image Upload

Field	Details
Test Scenario	Verify acceptance of valid images and rejection of invalid files
Test Steps	Upload JPG/PNG image → Start inspection → Repeat using an unsupported file
Test Data	Valid JPG/PNG; invalid TXT/EXE file
Expected Result	Supported image is accepted for inspection. Unsupported file is rejected with an appropriate message.
Actual Result	To be recorded
Pass/Fail Status	Not Executed

4.3 Fabric Defect Detection

TC-03 – Defect Detection, Classification and Location

Field	Details
Test Scenario	Verify identification of fabric defects
Test Steps	Upload a fabric image containing a known defect → Start inspection → View inspection result
Test Data	Fabric image containing a known defect
Expected Result	System detects the defect and identifies its type, severity, and location correctly.
Actual Result	To be recorded
Pass/Fail Status	Not Executed

4.4 Machine and Production Monitoring

TC-04 – Machine and Production Monitoring

Field	Details
Test Scenario	Verify real-time machine and production monitoring
Test Steps	Start machine → Monitor dashboard → Stop machine → Check production and downtime information
Test Data	Normal machine operation followed by machine stop
Expected Result	Machine status, production rate, and downtime are displayed and recorded correctly.
Actual Result	To be recorded
Pass/Fail Status	Not Executed

4.5 Analytics and Optimization

TC-05 – Root-Cause Analysis and Production Optimization

Field	Details
Test Scenario	Verify analysis of production data and optimization recommendations
Test Steps	Provide production and machine data → Run analysis → View recommendations
Test Data	Production data containing reduced output or increased defect rate
Expected	System identifies possible causes and provides appropriate production

Result	optimization recommendations.
Actual Result	To be recorded
Pass/Fail Status	Not Executed

4.6 Alert Generation

TC-06 – Defect Rate Threshold Testing

Field	Details
Test Scenario	Verify alert behavior below, at, and above the configured threshold
Test Steps	Enter defect rate below threshold → At threshold → Above threshold → Observe alerts
Test Data	Limit = 5%; Values = 4%, 5%, 6%
Expected Result	No alert below the threshold. The configured rule is applied at the boundary. An alert is generated when the threshold is exceeded.
Actual Result	To be recorded
Pass/Fail Status	Not Executed

4.7 Reporting and Data Management

TC-07 – Report Generation and Data Retrieval

Field	Details
Test Scenario	Verify generation and retrieval of production and quality reports
Test Steps	Select valid date range → Generate report → View stored production and inspection data
Test Data	Valid production records and date range
Expected Result	Report is generated correctly and relevant production and inspection data are retrieved accurately.
Actual Result	To be recorded
Pass/Fail Status	Not Executed

4.8 Exceptional Conditions

TC-08 – Sensor, Database and AI Service Failure

Field	Details
Test Scenario	Verify system behavior when critical services become unavailable
Test Steps	Disconnect sensor → Make database unavailable → Make AI service unavailable → Observe system
Test Data	Sensor, database, and AI service failure conditions
Expected Result	Failure is detected, an appropriate message or alert is displayed, and the system does not crash.
Actual Result	To be recorded
Pass/Fail Status	Not Executed

4.9 Performance and Usability

TC-09 – Performance and Dashboard Usability

Field	Details
Test Scenario	Verify dashboard response and usability during normal operation
Test Steps	Login → Open dashboard → View machine and production data → Navigate between major functions
Test Data	Normal production workload
Expected Result	Dashboard responds within the defined acceptable response time and information is clearly presented.
Actual Result	To be recorded
Pass/Fail Status	Not Executed

4.10 Scalability

TC-10 – Increased Workload and Connected Devices

Field	Details
Test Scenario	Verify system behavior when workload increases
Test Steps	Increase number of connected machines and sensor data → Monitor system performance
Test Data	Increased machines, users, and sensor data
Expected	System continues to process data correctly without unacceptable

Result	performance degradation or failure.
Actual Result	To be recorded
Pass/Fail Status	Not Executed

4.11 Test Case Summary

Test Area	Test Case
Authentication and Access Control	TC-01
Fabric Image Upload	TC-02
Defect Detection and Classification	TC-03
Machine and Production Monitoring	TC-04
Analytics and Optimization	TC-05
Alert Generation	TC-06
Reporting and Data Management	TC-07
Exceptional Conditions	TC-08
Performance and Usability	TC-09
Scalability	TC-10
Total	10 Test Cases

5. APPLICATION OF TEST DESIGN TECHNIQUES

5.1 Equivalence Partitioning

Equivalence Partitioning divides input data into valid and invalid groups so that representative values from each group can be tested.

Example: Fabric Image Upload

Partition	Test Data	Expected Result
Valid	JPG/PNG image	Image accepted
Invalid	TXT/EXE file	File rejected
Invalid	No file	Validation message displayed

TC-02 applies Equivalence Partitioning to verify different categories of image input.

5.2 Boundary Value Analysis

Boundary Value Analysis tests values at and around the limits of an accepted range.

Example: Defect Rate Threshold

The configured defect threshold is assumed to be 5%.

Test Value	Condition	Expected Result
4%	Below boundary	No alert
5%	Boundary value	Configured threshold rule is applied
6%	Above boundary	Alert generated

TC-06 applies Boundary Value Analysis to verify the alert threshold.

5.3 Decision Table Testing

Decision Table Testing verifies the system response for different combinations of conditions.

Defect Rate	Machine Status	Expected Alert
Below threshold	Running	No alert
Below threshold	Stopped	Machine-related alert
Above threshold	Running	Defect alert
Above threshold	Stopped	Defect and machine alert

This technique helps verify that alert behavior is consistent under different operating conditions.

5.4 State Transition Testing

State Transition Testing verifies how the system moves from one state to another when an event occurs.

Machine Monitoring Example

Running → Stopped → Maintenance → Running

Current State	Event	Expected Next State
Running	Machine failure	Stopped
Stopped	Maintenance started	Maintenance
Maintenance	Repair completed	Running

TC-04 and TC-08 can be used to verify machine and service state changes.

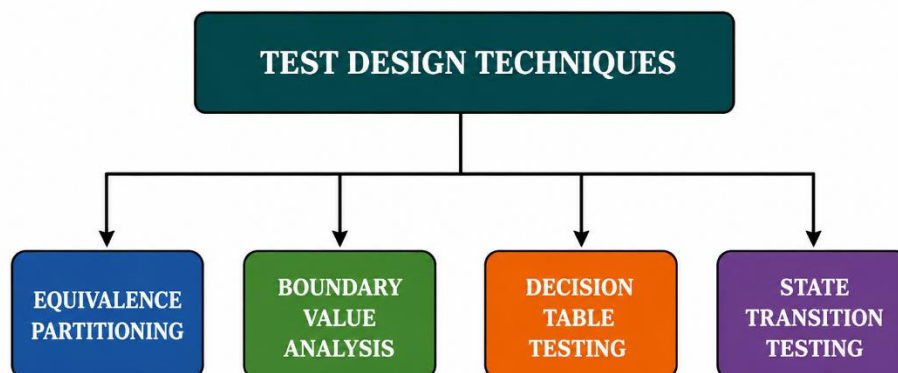


Figure 2 – Application of Test Design Techniques

6. REQUIREMENT TRACEABILITY AND TEST COVERAGE

6.1 Requirement Traceability Matrix

Requirement	Test Case / Scenario	Coverage
FR-01 – Authentication	TC-01	Covered
FR-02 – Access Management	TC-01	Covered
FR-03 – Image Upload	TC-02	Covered
FR-04 – Defect Detection	TC-03	Covered
FR-05 – Defect Classification	TC-03	Covered
FR-06 – Defect Location	TC-03	Covered
FR-07 – Machine Monitoring	TC-04	Covered
FR-08 – Production Monitoring	TC-04	Covered
FR-09 – Root-Cause Analysis	TC-05	Covered
FR-10 – Production Optimization	TC-05	Covered
FR-11 – Alert Generation	TC-06	Covered
FR-12 – Report Generation	TC-07	Covered
FR-13 – Data Storage/Retrieval	TC-07	Covered
FR-14 – Textile Process Support	TS-05	Scenario Covered
NFR-01 – Performance	TC-09	Covered
NFR-02 – Scalability	TC-10	Covered
NFR-03 – Security	TC-01	Covered
NFR-04 – Reliability	TC-08	Covered
NFR-05 – Availability	TC-08	Covered
NFR-06 – Maintainability	TC-08	Covered
NFR-07 – Usability	TC-09	Covered

6.2 Test Coverage

The test cases provide coverage for the identified functional and non-functional requirements.

The testing includes normal, invalid, boundary, and exceptional conditions.

The major coverage areas are:

- User authentication and access control
- Fabric image processing and defect inspection
- Machine and production monitoring
- Analytics and optimization
- Alert generation

- Reporting and data management
- Performance and usability
- Scalability
- Reliability and availability
- Exceptional system conditions

The traceability matrix connects requirements with corresponding test cases or test scenarios and helps ensure that important system functions are considered during testing.

7. CONCLUSION

The Test Case Design for the **Smart Textile Manufacturing Quality Inspection and Production Optimization Platform** covers the major functional and non-functional requirements of the proposed system.

Ten test cases have been designed to verify authentication, access control, fabric inspection, defect detection, machine monitoring, production analysis, alerts, reporting, system failures, performance, usability, and scalability.

The test cases include **normal, invalid, boundary, and exceptional conditions** to provide meaningful functional coverage.

Equivalence Partitioning, Boundary Value Analysis, Decision Table Testing, and State Transition Testing are applied to improve the effectiveness of the test design.

The Requirement Traceability Matrix establishes a clear relationship between requirements, test scenarios, and test cases. This provides a structured basis for evaluating the correctness, reliability, security, performance, availability, maintainability, and usability of the Smart Textile Manufacturing platform.